# Summary

**TABLE 1 Combinations and Permutations With and Without Repetition.**

| *Type* | *Repetition Allowed?* | *Formula* |
|---|---|---|
| $r$-permutations | No | $\frac{n!}{(n-r)!}$ |
| $r$-combinations | No | $\frac{n!}{r!\,(n-r)!}$ |
| $r$-permutations | Yes | $n^r$ |
| $r$-combinations | Yes | $\frac{(n+r-1)!}{r!\,(n-1)!}$ |